Scuola universitaria professionale
della Svizzera italiana

SUPSI

FSCLI

Software Engineering 2 - 2025/26

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Context & Motivation

- Application of **SE Principles**
  - use of **design patterns** and implementation of **tests**
- **O**bject **O**riented **P**rogramming
- Part of the **evaluation** of the SE2 course (5º semester)

Problem

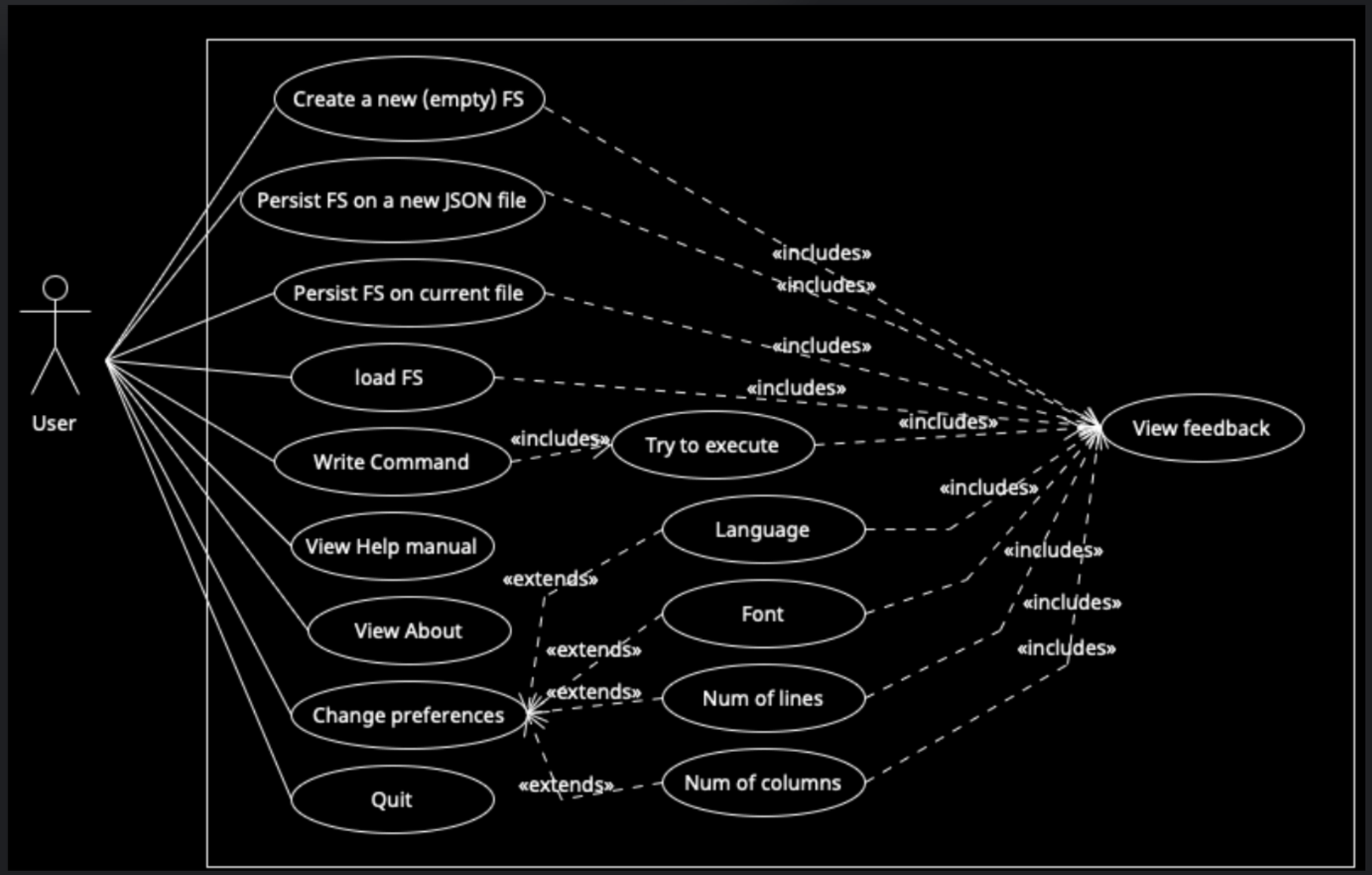
- Create a **virtual filesystem**
- **Commands** implementation and run, contextual feedback
- **Save**/**load** functionality
- Application **information** display -version number, build date, developers' credits
- **Standalone application**

State of Art

Not all of the above virtual FS allow the user to save/load files or modify preferencies.

- **Jimfs - Apache VFS** (Java)
- **Pyfakefs** (Python)
- **Afero** (Go)

Use case


User
Create a new (empty) FS
Persist FS on a new JSON file
Persist FS on current file
load FS
Write Command
View Help manual
View About
Change preferences
Quit
Try to execute
Language
Font
Num of lines
Num of columns
View feedback
«includes»
«extends»

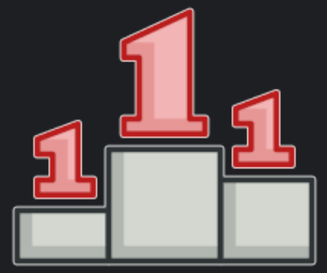
Approach

We faced the problem by focusing on:

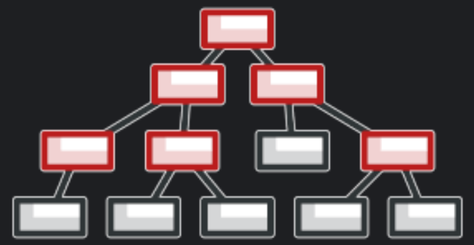
- Requirements Elicitation and Formalization
- Product versioning
- Software Design - Development - Tests
    - Patterns, 2 Modules (FE,BE), Packages
- Diagrams Design

Design Patterns - BE

Singleton



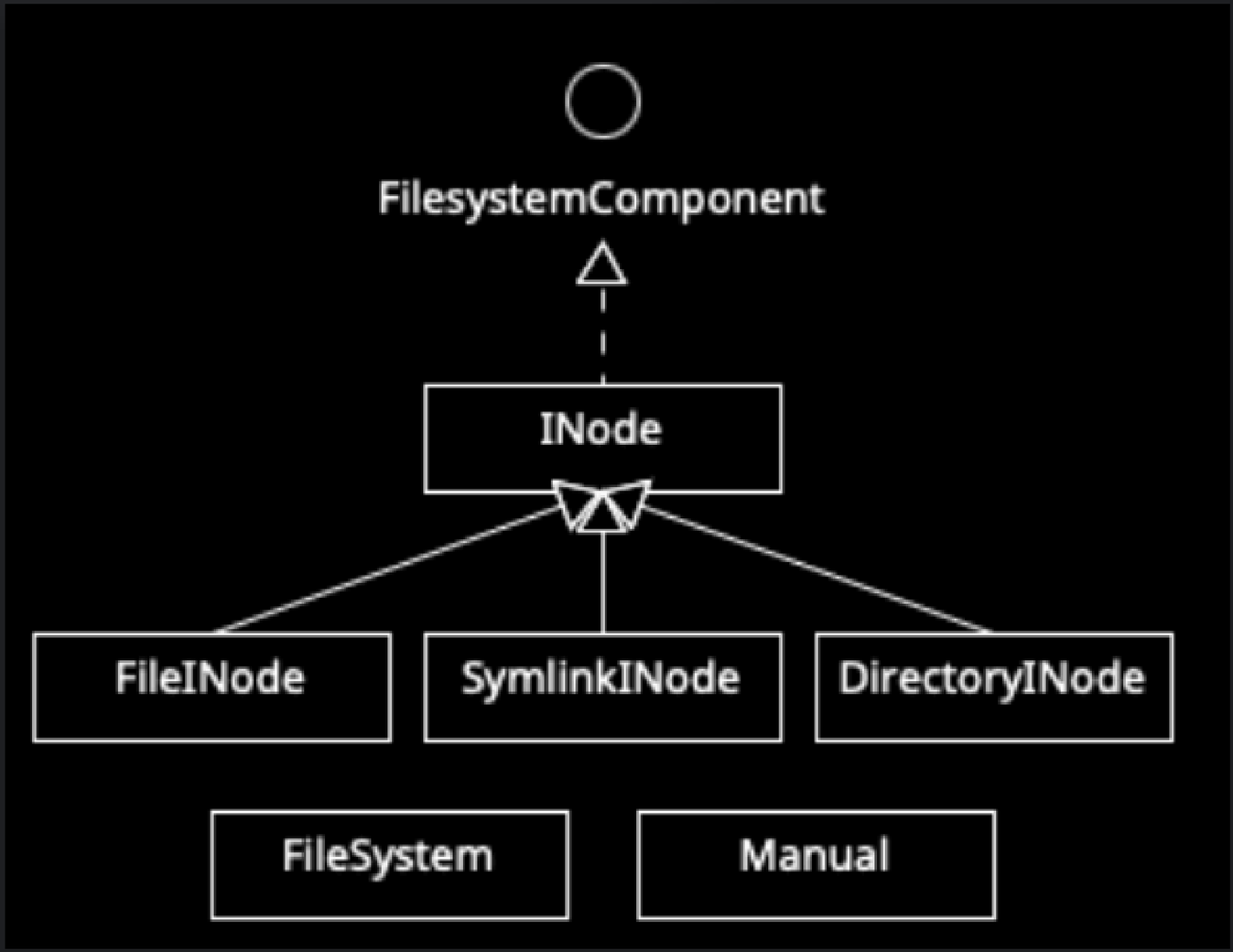
Composite



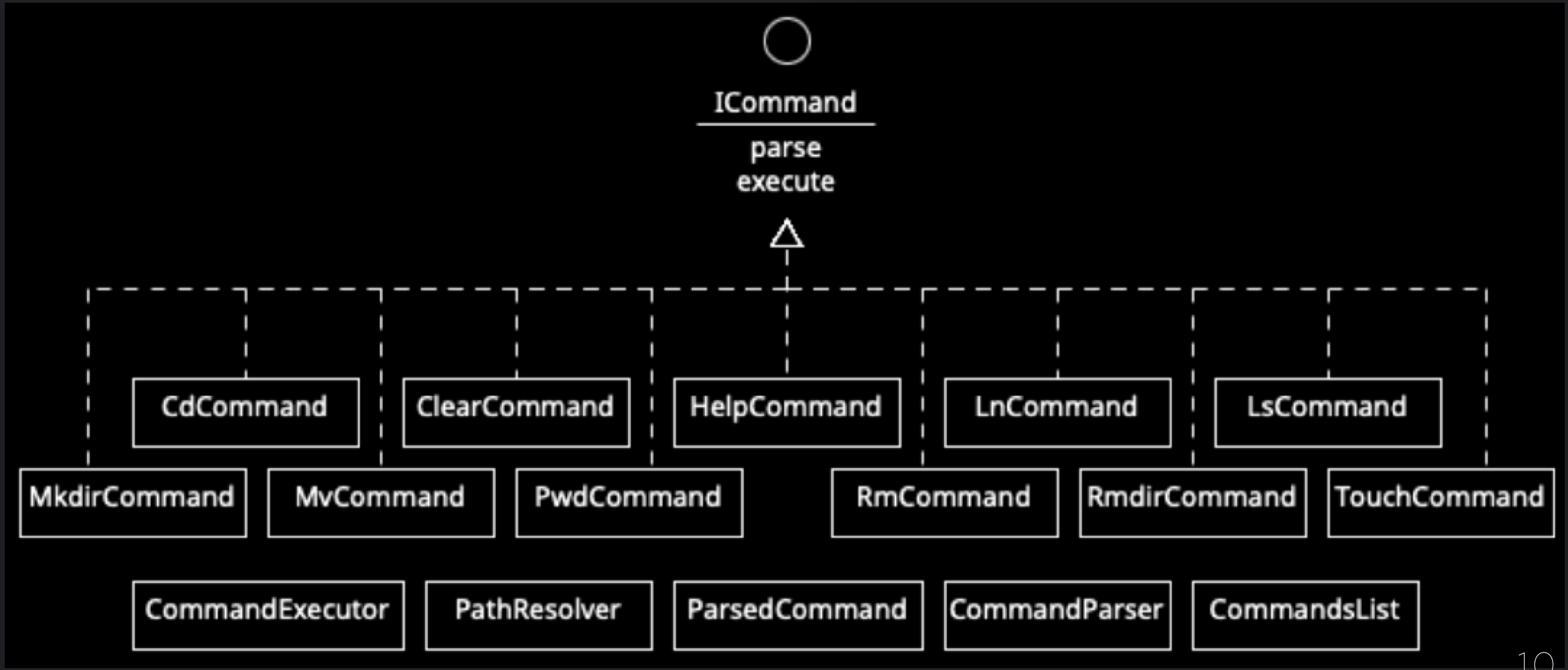
Command

Observer

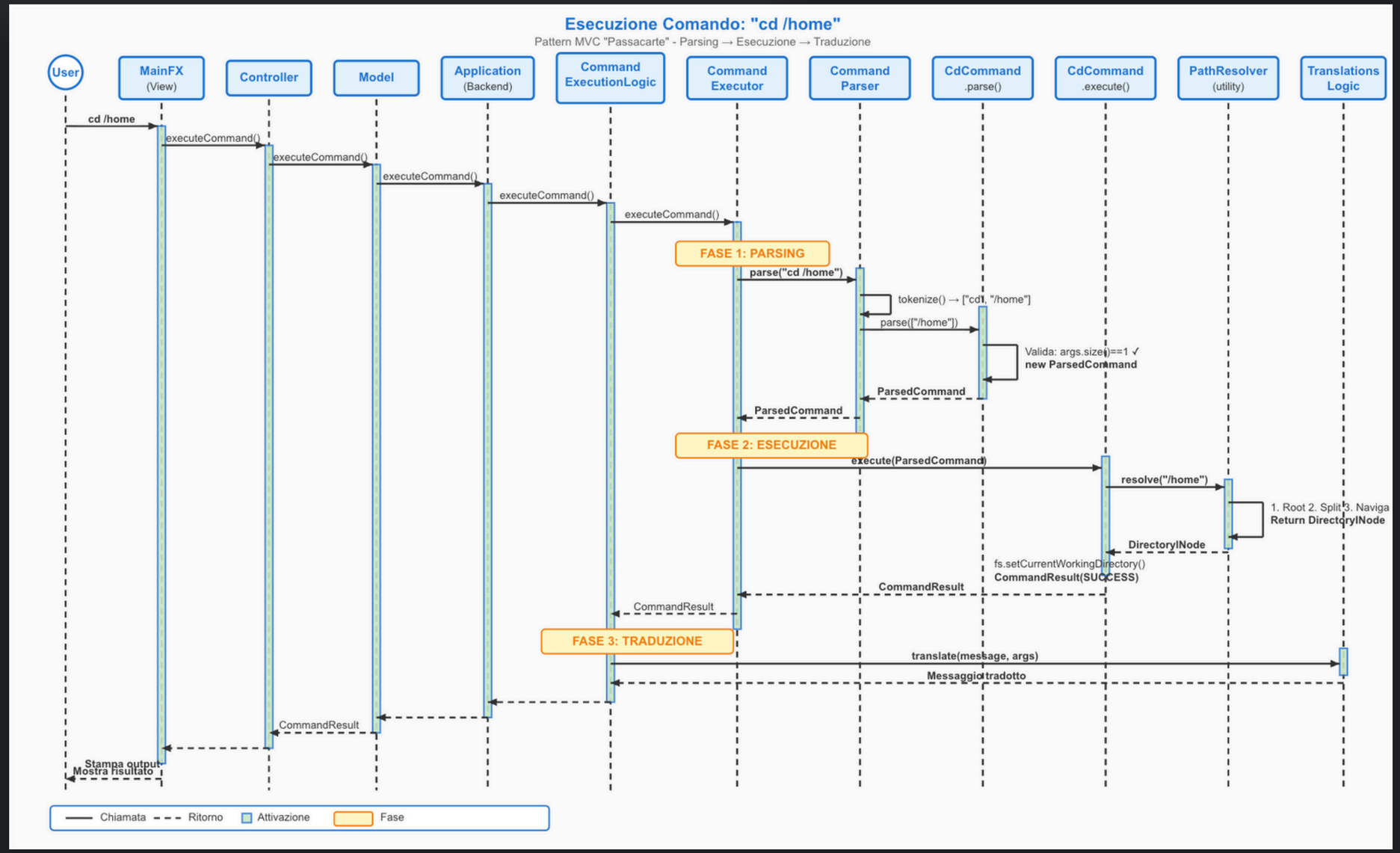
Class diagram pt.1



Class diagram pt.2



Sequence Diagram



DEMO

Results

We reached the following goals:

- ✓ **Completely working** Filesystem
- ✓ **Perstistence**
- ✓ Changable **preferences**
- ✓ Implementation of **patterns**

Conclusion

- All Requirements have been implemented.
- Efficient interaction with user

Difficult Parts:

- Fully understand commands and FS structure
- Patterns implementation
- E2E Tests implementation